```
import numpy as np
import matplotlib.pyplot as plt
```

```
hours = np.array([1,2,3,4,5,6,7,8,9,10])
scores = np.array([60,70,80,90,100,110,120,130,140,150])
```

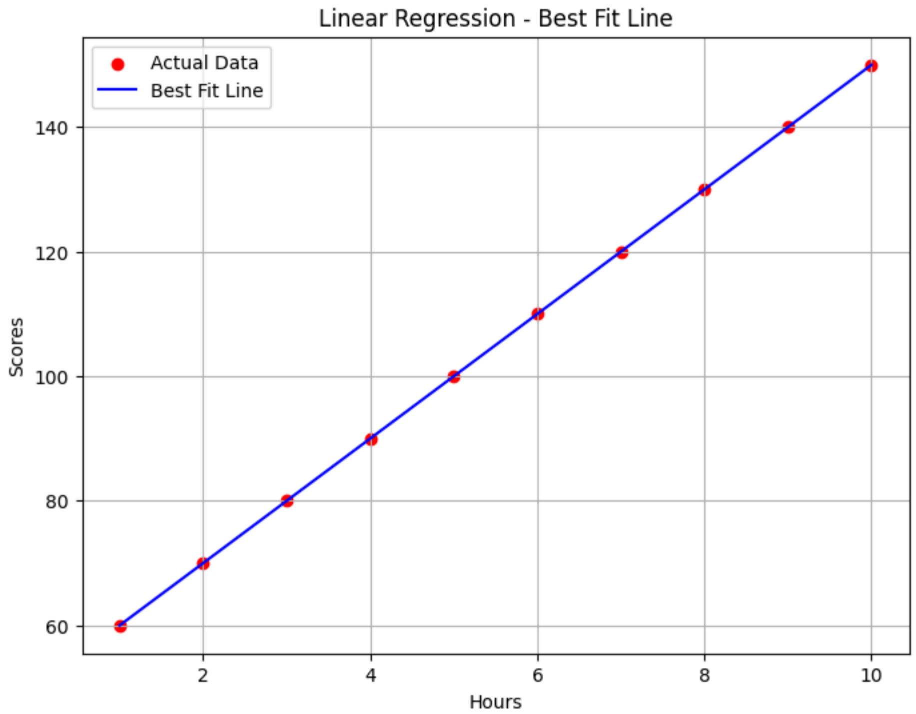
```
plt.figure(figsize=(8,6))

plt.scatter(hours, scores, label="Actual Data", color="red")
plt.plot(hours, predicted_scores, label="Best Fit Line", color="blue")

plt.xlabel("Hours")
plt.ylabel("Scores")
plt.title("Linear Regression - Best Fit Line")

plt.legend()
plt.grid(True)

plt.show()
```




```
predicted_score=10*5+50
print("Best Fit Line Equation:")
print("score = 10 * hours +50")
```

```
Best Fit Line Equation:
score = 10 * hours +50
```

```
print("\nFor 5 hours -> predicted score:",predicted_score)
```

```

For 5 hours -> predicted score: 100
```

```
import numpy as np
x=np.array([1,2,3,4,5],dtype=float)
y=np.array([60,70,80,90,100],dtype=float)
m=0
c=0
learning_rate=0.01
epochs=1000
n=len(x)
for i in range(epochs):
  y_pred=m*x+c
  error=y-y_pred
  mse=np.mean(error**2)
  dm=(-2/n)*np.sum(x*error)
  dc=(-2/n)*np.sum(error)
  m=m-(learning_rate*dm)
```

```
  c=c-(learning_rate*dc)
  if i%100==0:
    print(f"Epoch{i}")
    print(f"MSE={mse:.2f}")
    print(f"m{m:.2f},c={c:.2f}")
    print("------------------")
    print("\nFinal Best Fit Line:")
    print(f"y={m:.2f}x+{c:.2f}")
```

```
Epoch0
MSE=6600.00
m5.20,c=1.60
------------------

Final Best Fit Line:
y=5.20x+1.60
Epoch100
MSE=177.90
m18.63,c=18.84
------------------

Final Best Fit Line:
y=18.63x+18.84
Epoch200
MSE=90.37
m16.15,c=27.79
------------------

Final Best Fit Line:
y=16.15x+27.79
Epoch300
MSE=45.90
m14.38,c=34.17
------------------

Final Best Fit Line:
y=14.38x+34.17
Epoch400
MSE=23.32
m13.12,c=38.72
------------------

Final Best Fit Line:
y=13.12x+38.72
Epoch500
MSE=11.84
m12.23,c=41.96
------------------

Final Best Fit Line:
y=12.23x+41.96
Epoch600
MSE=6.02
m11.59,c=44.27
------------------

Final Best Fit Line:
y=11.59x+44.27
Epoch700
MSE=3.06
m11.13,c=45.92
------------------

Final Best Fit Line:
y=11.13x+45.92
Epoch800
MSE=1.55
```

Start coding or generate with AI.